

Watch-DevKit-HW

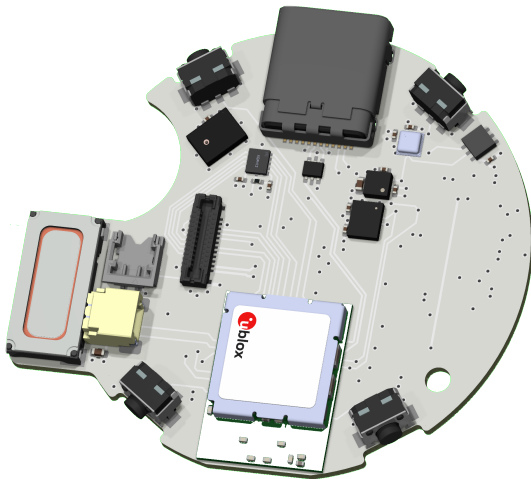
Variant: PRELIMINARY

2026-02-19

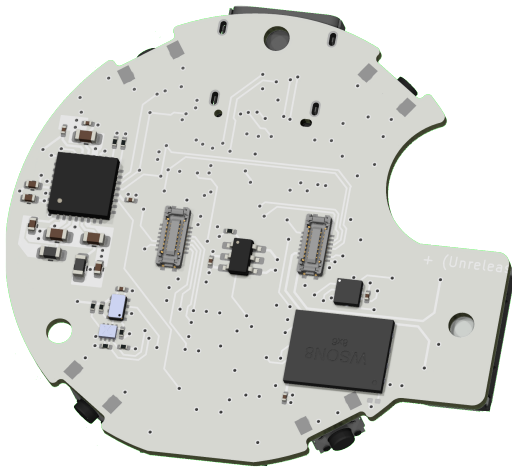
Rev + (Unreleased)

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TOP VIEW



BOTTOM VIEW



NOTES

Comment

Not fitted components are marked as **X**

DRAFT - Very early stage of schematic, ignore details.

PRELIMINARY - Close to final schematic.

CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.

RELEASED - A board with this schematic has been sent to production.

Date: 19-Feb-2026

DESIGN CONSIDERATIONS

DESIGN NOTE:

Example text for informational design notes.

DESIGN NOTE:

Example text for debug notes.

DESIGN NOTE:

Example text for cautionary design notes.

DESIGN NOTE:

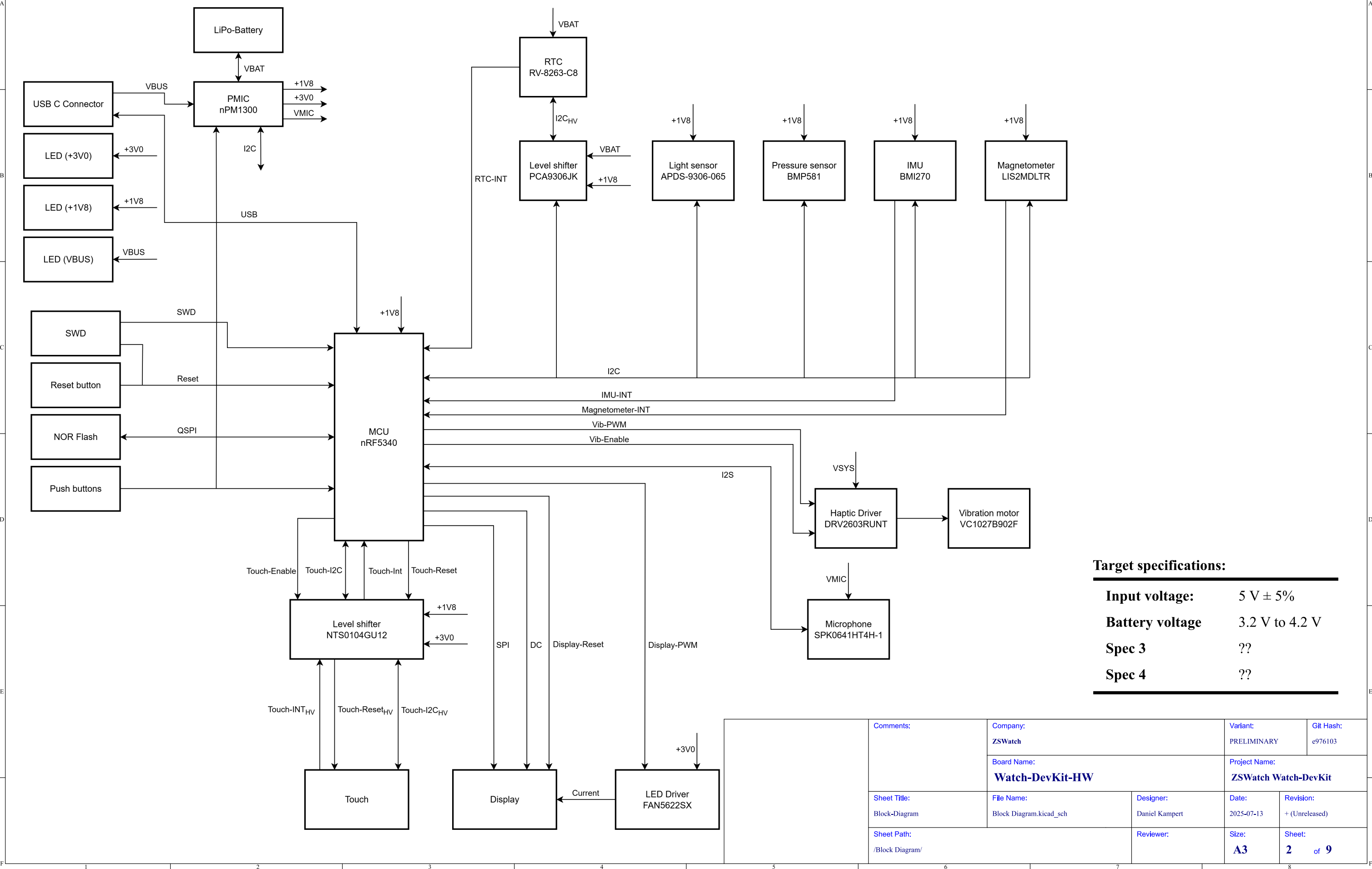
Example text for critical design notes.

LAYOUT NOTE:

Example text for critical layout guidelines.

	Comments:	Company: ZSWatch	Variant: PRELIMINARY	Git Hash: e976103
	Sheet Title: ZSWatch-Watch	Board Name: Watch-DevKit-HW	Project Name: ZSWatch Watch-DevKit	
	Sheet Path: /	File Name: ZSWatch-Watch.kicad_sch	Designer: Daniel Kampert	Date: 2025-07-13
			Reviewer:	Revision: + (Unreleased)
			Size: A3	Sheet: 1 of 9

[2] Block-Diagram



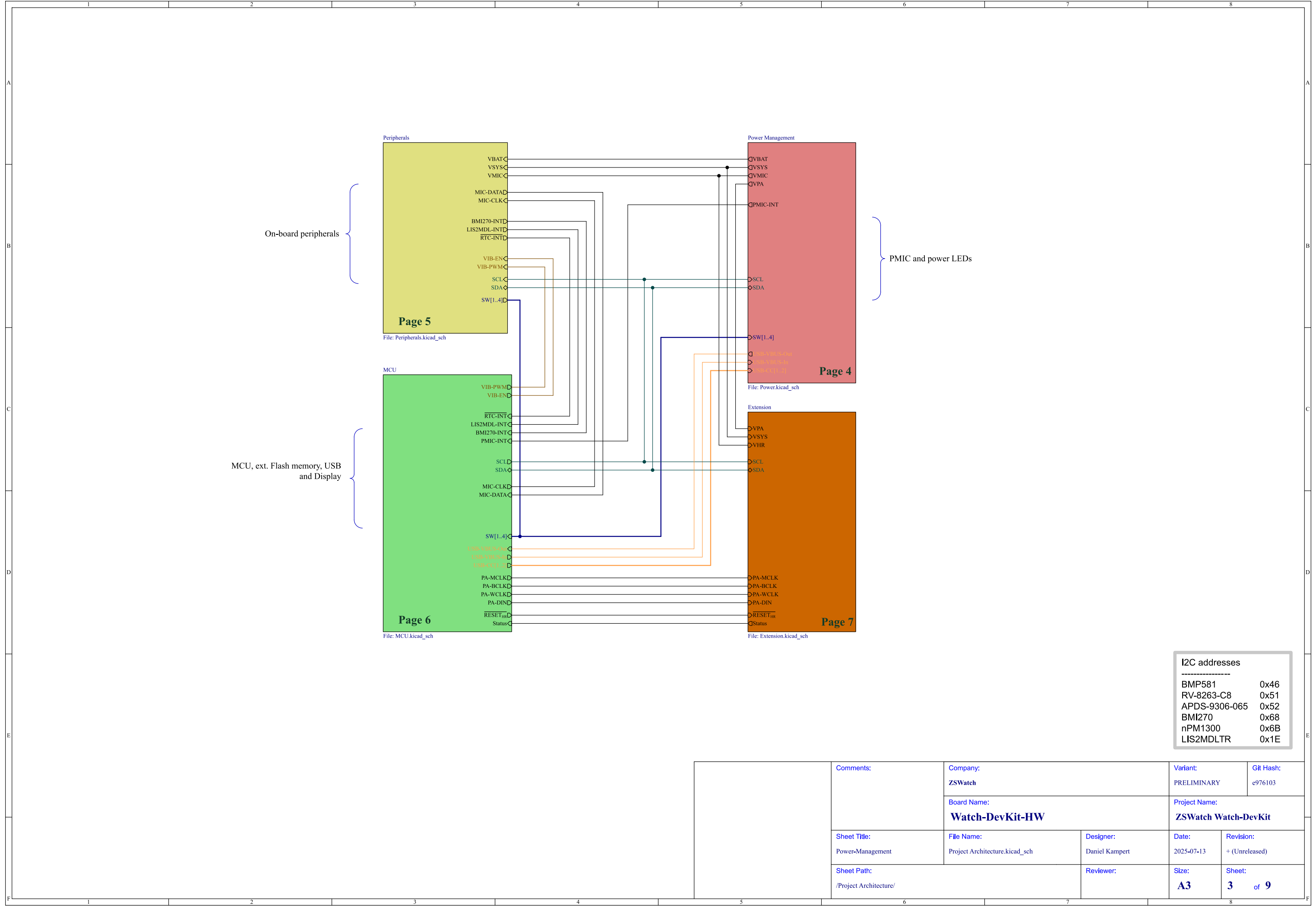
Target specifications:				
Input voltage:	5 V ± 5%			
Battery voltage	3.2 V to 4.2 V			
Spec 3	??			
Spec 4	??			

	Comments:	Company: ZSWatch		Variant: PRELIMINARY		Git Hash: e976103		
		Board Name: Watch-DevKit-HW		Project Name: ZSWatch Watch-DevKit				
	Sheet Title: Block-Diagram	File Name: Block Diagram.kicad_sch		Designer: Daniel Kampert		Date: 2025-07-13	Revision: + (Unreleased)	
	Sheet Path: /Block Diagram/			Reviewer:		Size: A3		Sheet: 2 of 9

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

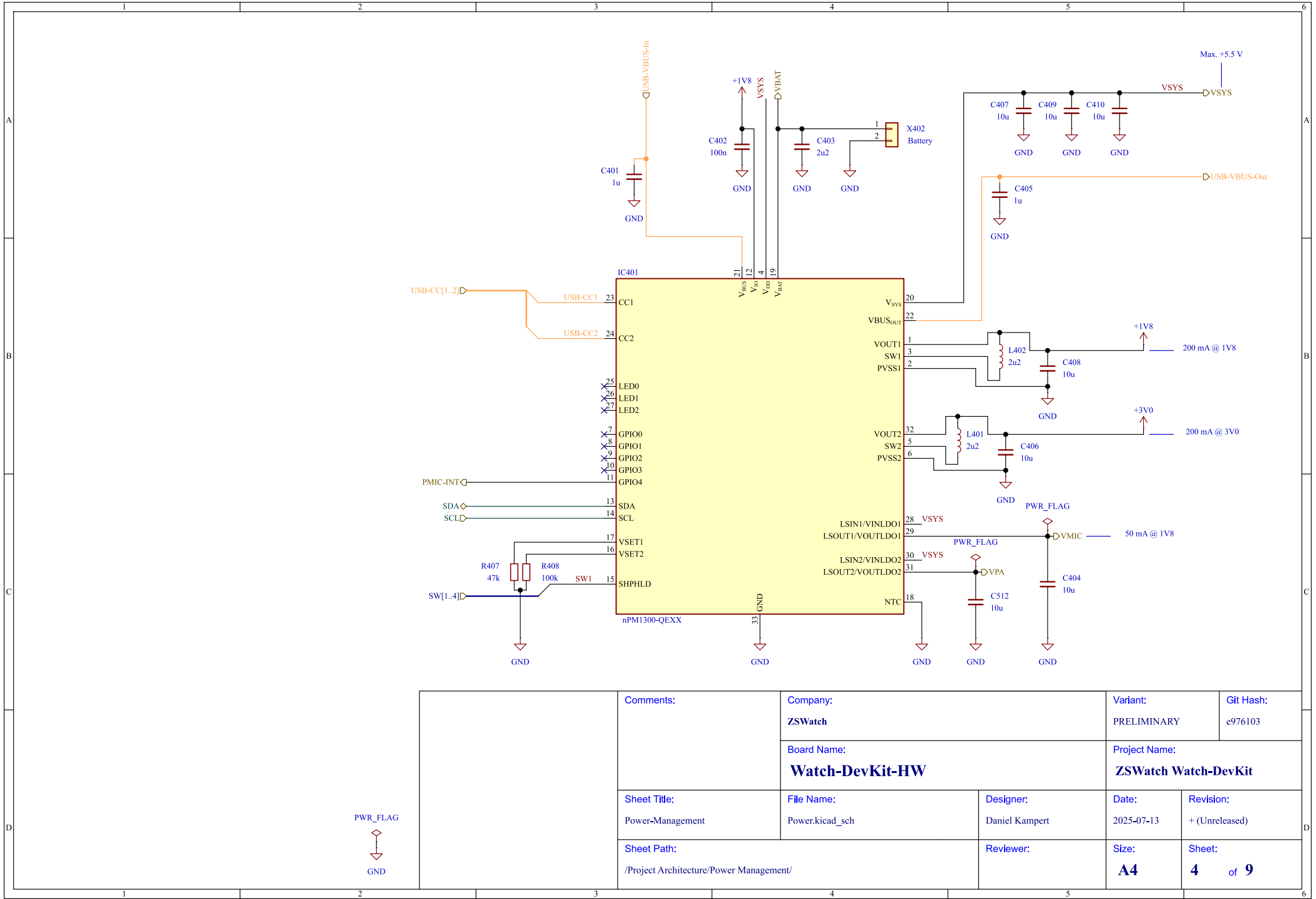
Target specifications:	
Input voltage:	?? - ?? V
Spec 2	??
Spec 3	??
Spec 4	??

	Comments:	Company: ZSWatch		Variant: PRELIMINARY	Git Hash: e976103
		Board Name: Watch-DevKit-HW		Project Name: ZSWatch Watch-DevKit	
	Sheet Title:	File Name: Power - Topology.kicad_sch	Designer: Daniel Kampert	Date: 2025-07-13	Revision: + (Unreleased)
	Sheet Path: /Power - Topology/		Reviewer:	Size: A3	Sheet: 2 of 9

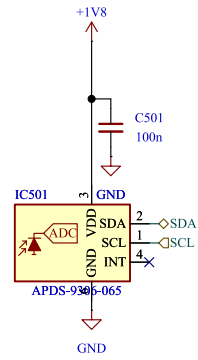


I2C addresses	
BMP581	0x46
RV-8263-C8	0x51
APDS-9306-065	0x52
BMI270	0x68
nPM1300	0x6B
LIS2MDLTR	0x1E

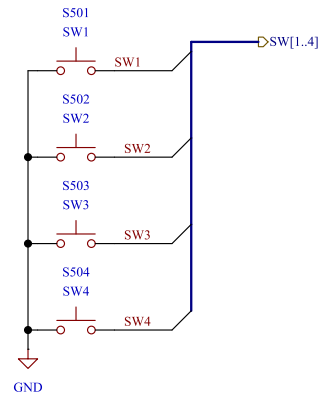
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	Sheet Title: Power-Management	Board Name: Watch-DevKit-HW		Project Name: ZSWatch Watch-DevKit	
	Sheet Path: /Project Architecture/	File Name: Project Architecture.kicad_sch	Designer: Daniel Kampert	Date: 2025-07-13	Revision: + (Unreleased)
			Reviewer:	Size: A3	Sheet: 3 of 9



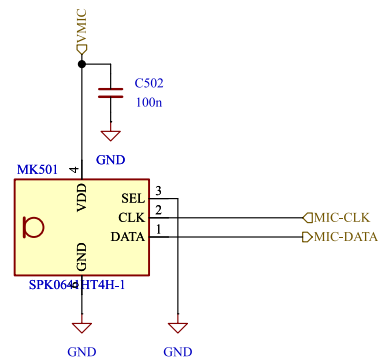
Light sensor



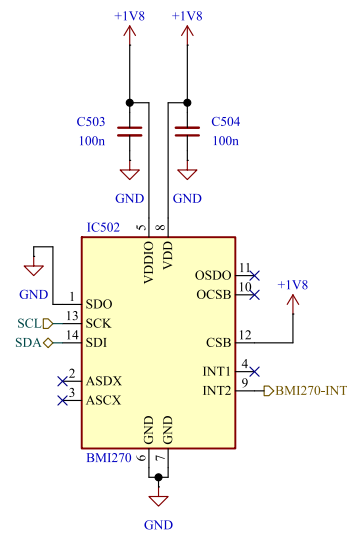
Buttons



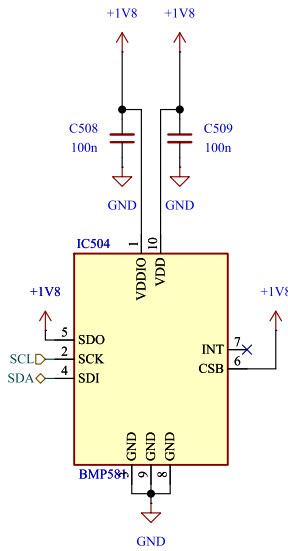
Microphone



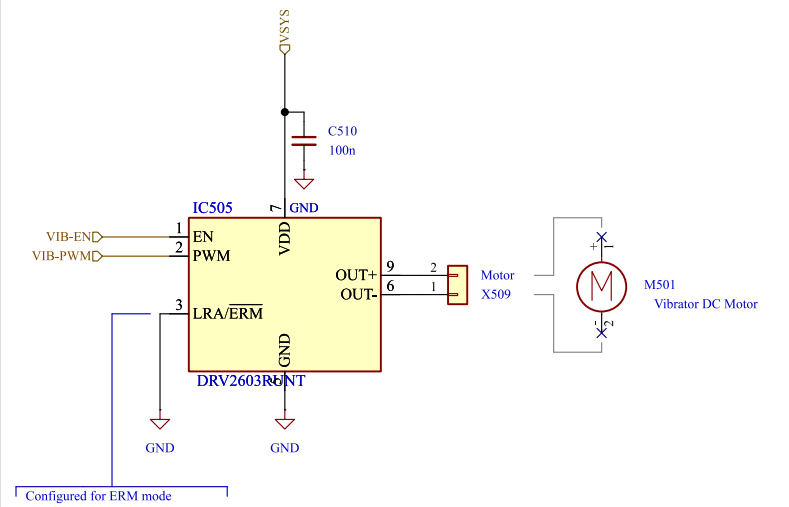
IMU with step counter and gesture detection



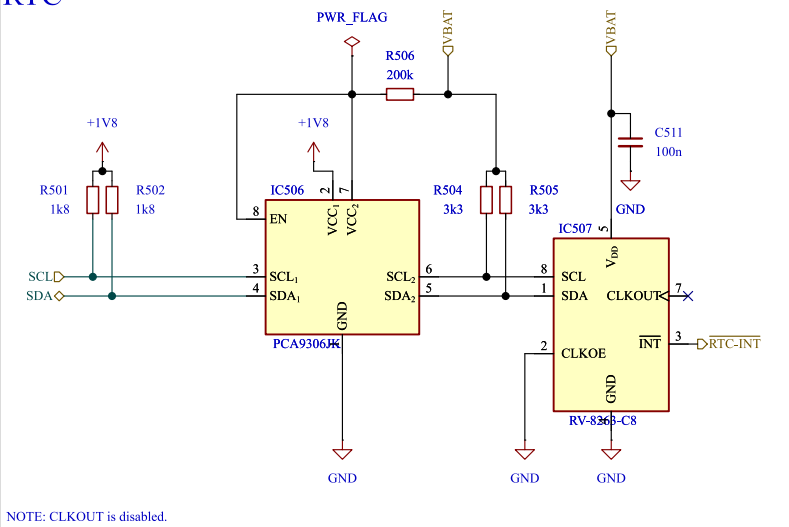
High accuracy pressure sensor



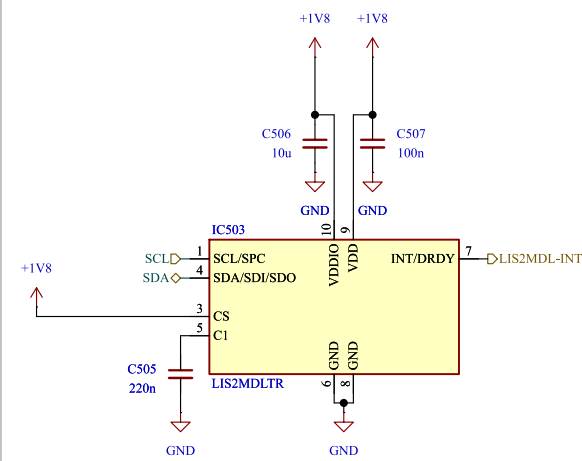
Vibration motor



RTC

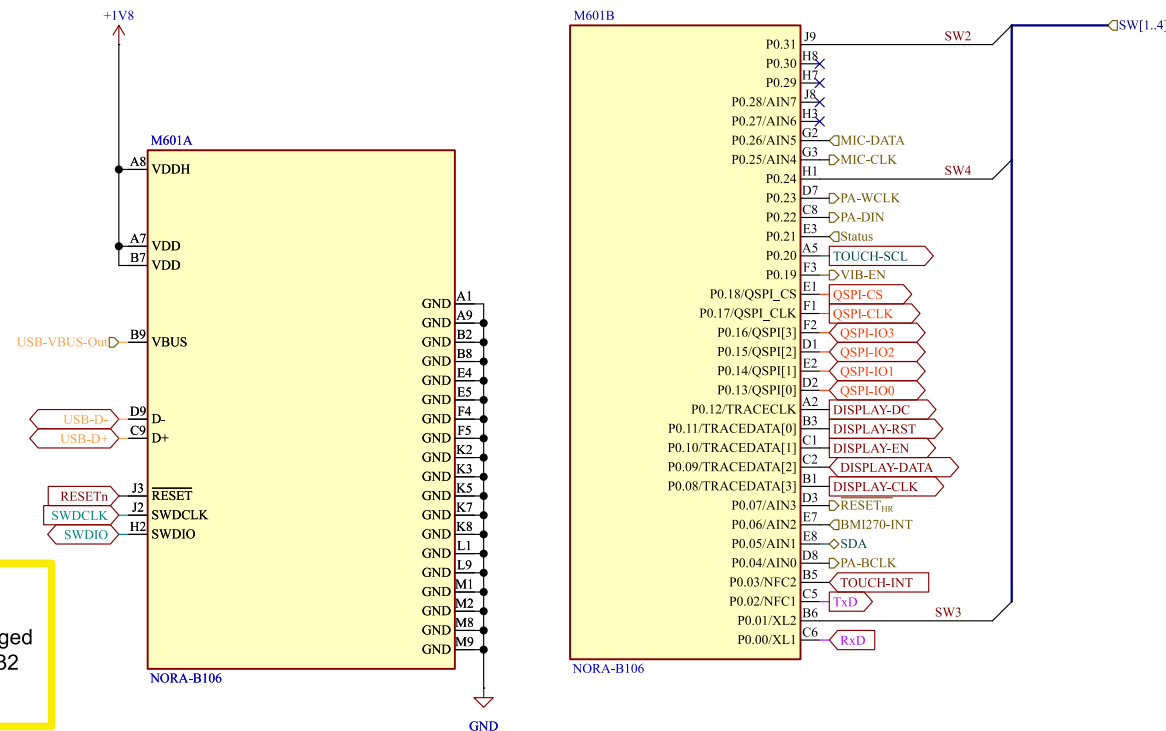


Magnetometer



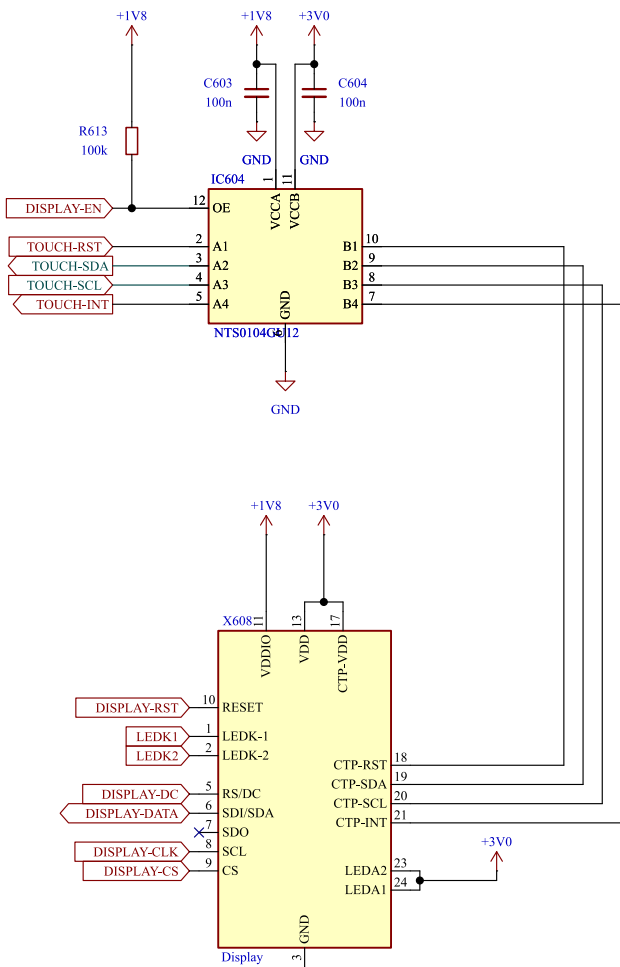
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	Board Name: Watch-DevKit-HW		Project Name: ZSWatch Watch-DevKit	
Sheet Title: Peripherals	File Name: Peripherals.kicad_sch	Designer: Daniel Kampert	Date: 2025-07-13	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Peripherals/		Reviewer:	Size: A3	Sheet: 5 of 9

Microcontroller
(NORA-B106/nRF5340)



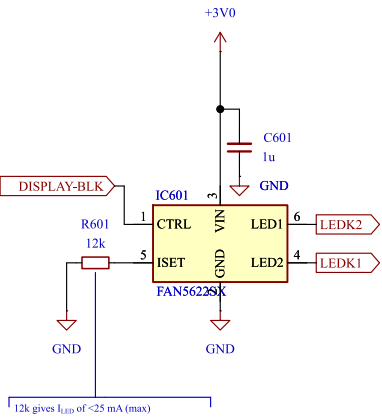
DESIGN NOTE:
The SPI lines for the display can't be changed and are required for 32 MHz operation.

Display

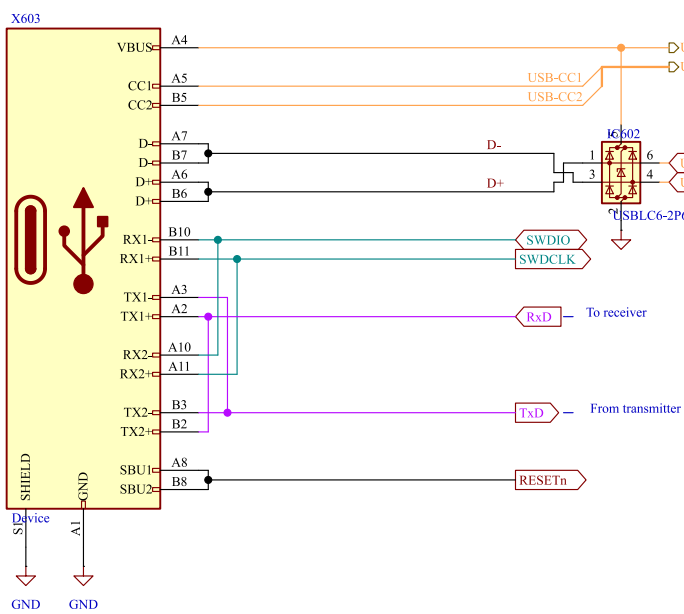


DESIGN NOTE:
NTS0104GU12 has internal pullups.

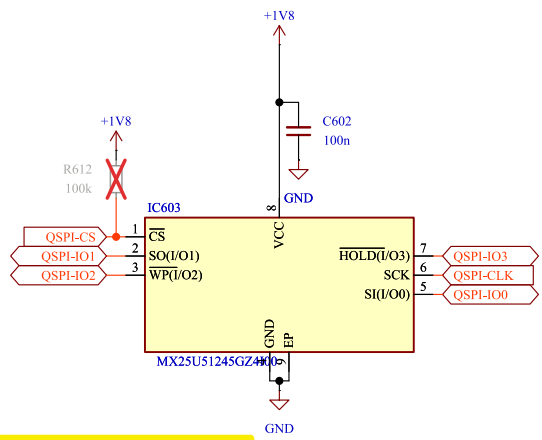
Display backlight



USB

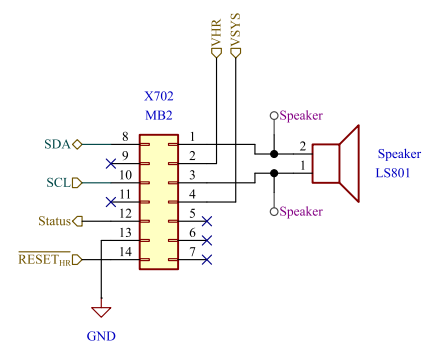
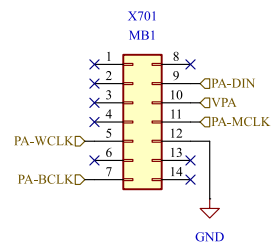


External Flash



DESIGN NOTE:
Make sure that the Flash Memory supports 4k sector erase commands (limited by QSPI driver)!

Comments:	Company: ZSWatch		Variant: PRELIMINARY		Git Hash: e976103	
	Board Name: Watch-DevKit-HW		Project Name: ZSWatch Watch-DevKit			
Sheet Title: MCU	File Name: MCU.kicad_sch	Designer: Daniel Kampert	Date: 2025-07-13	Revision: + (Unreleased)		
Sheet Path: /Project Architecture/MCU/		Reviewer:	Size: A3	Sheet: 6 of 9		



	Comments:	Company: ZSWatch		Variant: PRELIMINARY	Git Hash: e976103
		Board Name: Watch-DevKit-HW		Project Name: ZSWatch Watch-DevKit	
	Sheet Title: Extension	File Name: Extension.kicad_sch	Designer: Daniel Kampert	Date: 2025-07-13	Revision: + (Unreleased)
	Sheet Path: /Project Architecture/Extension/		Reviewer:	Size: A4	Sheet: 7 of 9

[8] Revision-History

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		Board Name: Watch-DevKit-HW		Project Name: ZSWatch Watch-DevKit	
	Sheet Title: Revision-History	File Name: Revision History.kicad_sch	Designer: Daniel Kampert	Date: 2025-07-13	Revision: + (Unreleased)
	Sheet Path: /Revision History/		Reviewer:	Size: A4	Sheet: 8 of 9